Zhaoyuan Wan

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Research Interests: Embodied AI · 3D Computer Vision · Machine Learning Systems

Education

Shanghai Jiao Tong University (SJTU) Sept 2023 – Present

B.Eng. in Computer Science (ACM Honors Class), Zhiyuan College **Rank: 2 / 26 · GPA: 4.12 / 4.30 (94.49 / 100)**

- **Honors:** National Scholarship (2024, Top 0.2% nationwide), SJTU Zhiyuan Honors Scholarship (2023).
- **CS Coursework:** Computer Architecture (100), Compiler Design (100), Programming in C++ Honors (98), Discrete Mathematics Honors (98), Data Structures Honors (97).
- **Math Coursework:** Mathematical Analysis Honors II (97), Probability & Statistics Honors (97), Linear Algebra Honors (96), Mathematical Methods in Physics Honors (96), Mathematical Analysis Honors I (95).

Research Experience

MVIG Lab, Shanghai Jiao Tong University & Noematrix July 2026 – Present

Research Intern · Advised by Prof. Cewu Lu

- Researching pre-training and post-training of World Action Models (WAM) and Vision-Language-Action (VLA) policies for generalizable robotic manipulation.
- Investigating 3D geometric transition distillation and spatial feature alignment to improve multi-view manipulation generalization across simulation (LIBERO, RoboTwin 2.0) and real-world platforms.

Selected Projects

STEVE: Single-Image 3D Minecraft Architectural Reconstruction *PyTorch, 3D Vision, VQ-VAE* · 2026

- Designed an end-to-end deep learning pipeline reconstructing 3D Minecraft structures (`.schem`) from a single screenshot.
- Combined a multi-view VQ-VAE block tokenizer, a U-Net monocular depth and camera pose estimator, and a frustum-backprojected 3D CNN for occupancy and block token prediction.

POLARIS: OS-Level Paged KV-Cache Layer for LLM Inference *Linux Kernel (C), Rust, CUDA VMM* · 2026

- Co-designed a Linux kernel module (`polaris.ko`) and a Rust userspace daemon (`polarisd`) for OS-managed paged attention memory.
- Orchestrated demand paging, asynchronous GPU↔CPU swapping, and copy-on-write (CoW) prefix sharing via CUDA Virtual Memory Management (VMM) APIs.

R10K-CPU & c0mpiler: Out-of-Order RISC-V Processor & Optimizing Rust Compiler *SystemVerilog, Rust, LLVM* · 2026

- Implemented an out-of-order RV32IM processor modeled after the MIPS R10K microarchitecture, featuring a 64-entry physical register file, register renaming, active list (ROB), non-blocking LSQ, SRT divider, and branch prediction.
- Built an end-to-end optimizing compiler (`c0mpiler`) from a Rust subset to LLVM IR and RV32IM assembly in Rust, with SSA construction, dataflow optimizations, and graph-coloring register allocation.

Monte Carlo Ray Tracer *Rust, Computer Graphics* · 2025

- Built a physically-based path tracer from scratch in Rust featuring Probability Density Function (PDF) mixture importance sampling, full Disney Principled BSDF, portal light transport, and BVH acceleration.

Teaching Experience

Teaching Assistant, SJTU ACM Honors Class Fall 2026

- **CS2966 Compiler Design:** Authored the formal *Rx Language Specification* (defining the Rust-subset EBNF grammar, type system, and runtime semantics) and curated the compiler test suite.
- **CS2967 Computer Architecture:** Designed the Out-of-Order RV32IM CPU grading framework with C++ AXI4-Lite memory co-simulation, Verilator IPC benchmarking, and Yosys + ASAP7 7nm area synthesis.

Technical Skills

- **Languages & Hardware:** Rust, C/C++, Python, SystemVerilog, RISC-V Architecture.
- **Systems & Machine Learning:** PyTorch, CUDA, LLVM, Linux Kernel Development, 3D Vision & Embodied AI Toolchains.